

SKILL(WEEK-3) DSEDBD SEC-10

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1. Create Tables for Joins

The screenshot shows the SQL Developer interface with the 'SQL File 3*' editor open. The editor contains the following SQL code:

```
1 • create database JOINS_AND_SETS;
2 • use JOINS_AND_SETS;
3
4 • CREATE TABLE class (
5     id INT,
6     name VARCHAR(30)
7 );
8 • CREATE TABLE class_info (
9     id INT,
10    address VARCHAR(30)
11 );
```

The Navigator pane on the left shows the 'bookflow_db' schema selected. The Output pane at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	08:53:24	create database JOINS_AND_SETS	1 row(s) affected	0.047 sec
2	08:53:42	use JOINS_AND_SETS	0 row(s) affected	0.000 sec
3	08:53:50	CREATE TABLE class (id INT, name ...	0 row(s) affected	0.078 sec
4	08:54:02	CREATE TABLE class_info (id INT, a...	0 row(s) affected	0.047 sec

2. Insert Data for Cross Join

The screenshot shows the SQL Developer interface with the 'SQL File 3*' editor open. The editor contains the following SQL code:

```
44
45 • CREATE TABLE class_info (
46     id INT,
47     address VARCHAR(30)
48 );
```

The Navigator pane on the left shows the 'bookflow_db' schema selected. The Output pane at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
4	09:06:36	USE JOINS_AND_SETS	0 row(s) affected	0.000 sec
5	09:06:39	CREATE TABLE class_info (id INT, ...	0 row(s) affected	0.031 sec
6	09:07:02	INSERT INTO class_info VALUES (1, 'D...	3 row(s) affected Records: 3 Duplicates...	0.016 sec
7	09:07:23	SELECT * FROM class LIMIT 0, 1000	Error Code: 1146. Table joins_and_sets...	0.000 sec
8	09:09:08	INSERT INTO class_info VALUES (1, 'D...	3 row(s) affected Records: 3 Duplicates...	0.016 sec
9	09:09:23	USE JOINS_AND_SETS	0 row(s) affected	0.016 sec
10	09:09:33	SELECT * FROM class LIMIT 0, 1000	Error Code: 1146. Table joins_and_sets...	0.000 sec
11	09:10:00	CREATE DATABASE IF NOT EXISTS J...	1 row(s) affected, 1 warning(s): 1007 Can...	0.000 sec
12	09:10:26	USE JOINS_AND_SETS	0 row(s) affected	0.016 sec
13	09:10:34	CREATE TABLE class (id INT, na...	0 row(s) affected	0.031 sec
14	09:10:45	SHOW TABLES	2 row(s) returned	0.000 sec / 0.000 sec
15	09:11:02	INSERT INTO class VALUES (1, 'abhi')...	4 row(s) affected Records: 4 Duplicates...	0.000 sec
16	09:11:10	SELECT * FROM class LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

root x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- 25r21a05c9
- bankdb
- bookflow_db
- sys

SQL File 3* x

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

50 • INSERT INTO class_info VALUES
51 (1, 'DELHI'),
52 (2, 'MUMBAI'),
53 (3, 'CHENNAI');
54 • SELECT * FROM class_info;

```

Result Grid

	id	address
1	DELHI	
2	MUMBAI	
3	CHENNAI	
1	DELHI	
2	MUMBAI	
3	CHENNAI	

class_info 3 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	09:07:23	SELECT * FROM class LIMIT 0, 1000	Error Code: 1146. Table 'joins_and_sets'...	0.000 sec
8	09:09:08	INSERT INTO class_info VALUES (1, 'D...	3 row(s) affected Records: 3 Duplicates:...	0.016 sec
9	09:09:23	USE JOINS_AND_SETS	0 row(s) affected	0.016 sec
10	09:09:33	SELECT * FROM class LIMIT 0, 1000	Error Code: 1146. Table 'joins_and_sets'...	0.000 sec
11	09:10:00	CREATE DATABASE IF NOT EXISTS J...	1 row(s) affected, 1 warning(s): 1007 Can...	0.000 sec
12	09:10:26	USE JOINS_AND_SETS	0 row(s) affected	0.016 sec
13	09:10:34	CREATE TABLE class (id INT, na...	0 row(s) affected	0.031 sec
14	09:10:45	SHOW TABLES	2 row(s) returned	0.000 sec / 0.000 sec
15	09:11:02	INSERT INTO class VALUES (1, 'abhi'), ...	4 row(s) affected Records: 4 Duplicates:...	0.000 sec
16	09:11:10	SELECT * FROM class LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
17	09:12:04	CREATE TABLE class_info (id INT, ...	Error Code: 1050. Table 'class_info' alrea...	0.000 sec
18	09:12:20	INSERT INTO class_info VALUES (1, 'D...	3 row(s) affected Records: 3 Duplicates:...	0.031 sec
19	09:12:30	SELECT * FROM class_info LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec

Administration Schemas

Information

No object selected

Object Info Session

3. Cross Join Query

root x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- 25r21a05c9
- bankdb
- bookflow_db
- sys

SQL File 3* x

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

54 • SELECT * FROM class_info;
55
56 • SELECT *
57 FROM class
58 CROSS JOIN class_info;

```

Result Grid

	id	name	id	address
4	anu	1	DELHI	
3	alex	1	DELHI	
2	adam	1	DELHI	
1	abhi	1	DELHI	
4	anu	2	MUMBAI	
3	alex	2	MUMBAI	
2	adam	2	MUMBAI	
1	abhi	2	MUMBAI	

Result 4 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	09:09:08	INSERT INTO class_info VALUES (1, 'D...	3 row(s) affected Records: 3 Duplicates:...	0.016 sec
9	09:09:23	USE JOINS_AND_SETS	0 row(s) affected	0.016 sec
10	09:09:33	SELECT * FROM class LIMIT 0, 1000	Error Code: 1146. Table 'joins_and_sets'...	0.000 sec
11	09:10:00	CREATE DATABASE IF NOT EXISTS J...	1 row(s) affected, 1 warning(s): 1007 Can...	0.000 sec
12	09:10:26	USE JOINS_AND_SETS	0 row(s) affected	0.016 sec
13	09:10:34	CREATE TABLE class (id INT, na...	0 row(s) affected	0.031 sec
14	09:10:45	SHOW TABLES	2 row(s) returned	0.000 sec / 0.000 sec
15	09:11:02	INSERT INTO class VALUES (1, 'abhi'), ...	4 row(s) affected Records: 4 Duplicates:...	0.000 sec
16	09:11:10	SELECT * FROM class LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
17	09:12:04	CREATE TABLE class_info (id INT, ...	Error Code: 1050. Table 'class_info' alrea...	0.000 sec
18	09:12:20	INSERT INTO class_info VALUES (1, 'D...	3 row(s) affected Records: 3 Duplicates:...	0.031 sec
19	09:12:30	SELECT * FROM class_info LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
20	09:12:50	SELECT * FROM class CROSS JOIN cla...	36 row(s) returned	0.000 sec / 0.000 sec

Administration Schemas

Information

No object selected

Object Info Session

4. Reset Data for Main Joins

The screenshot shows the SQL Developer interface with a SQL query in the main editor. The query is:

```
69 ON class.id = class_info.id;
70
71 SELECT *
72 FROM class
73 NATURAL JOIN class_info;
```

The result grid displays the following data:

	id	name	address
1	abhi	DELHI	
2	adam	MUMBAI	
3	alex	CHENNAI	
1	abhi	DELHI	
2	adam	MUMBAI	
3	alex	CHENNAI	

The output pane shows the following actions:

#	Time	Action	Message	Duration / Fetch
11	09:10:00	CREATE DATABASE IF NOT EXISTS J...	1 row(s) affected, 1 warning(s): 1007 Can...	0.000 sec
12	09:10:26	USE JOINS_AND_SETS	0 row(s) affected	0.016 sec
13	09:10:34	CREATE TABLE class (id INT, na...	0 row(s) affected	0.031 sec
14	09:10:45	SHOW TABLES	2 row(s) returned	0.000 sec / 0.000 sec
15	09:11:02	INSERT INTO class VALUES (1, 'abhi'), ...	4 row(s) affected Records: 4 Duplicates:...	0.000 sec
16	09:11:10	SELECT * FROM class LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
17	09:12:04	CREATE TABLE class_info (id INT, ...	Error Code: 1050. Table 'class_info' alrea...	0.000 sec
18	09:12:20	INSERT INTO class_info VALUES (1, 'D...	3 row(s) affected Records: 3 Duplicates:...	0.031 sec
19	09:12:30	SELECT * FROM class_info LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
20	09:12:50	SELECT * FROM class CROSS JOIN cla...	36 row(s) returned	0.000 sec / 0.000 sec
21	09:13:58	SELECT * FROM class INNER JOIN cla...	9 row(s) returned	0.000 sec / 0.000 sec
22	09:14:18	SELECT class.name, class_info.add...	9 row(s) returned	0.000 sec / 0.000 sec
23	09:15:32	SELECT * FROM class NATURAL JOIN ...	9 row(s) returned	0.000 sec / 0.000 sec

5. Inner Join Queries

The screenshot shows the SQL Developer interface with a SQL query in the main editor. The query is:

```
81
82 SELECT *
83 FROM class
84 LEFT OUTER JOIN class_info
85 ON class.id = class_info.id;
```

The result grid displays the following data:

	id	name	id	address
1	abhi	1	DELHI	
1	abhi	1	DELHI	
1	abhi	1	DELHI	
2	adam	2	MUMBAI	
2	adam	2	MUMBAI	
2	adam	2	MUMBAI	

The output pane shows the following actions:

#	Time	Action	Message	Duration / Fetch
13	09:10:34	CREATE TABLE class (id INT, na...	0 row(s) affected	0.031 sec
14	09:10:45	SHOW TABLES	2 row(s) returned	0.000 sec / 0.000 sec
15	09:11:02	INSERT INTO class VALUES (1, 'abhi'), ...	4 row(s) affected Records: 4 Duplicates:...	0.000 sec
16	09:11:10	SELECT * FROM class LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
17	09:12:04	CREATE TABLE class_info (id INT, ...	Error Code: 1050. Table 'class_info' alrea...	0.000 sec
18	09:12:20	INSERT INTO class_info VALUES (1, 'D...	3 row(s) affected Records: 3 Duplicates:...	0.031 sec
19	09:12:30	SELECT * FROM class_info LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
20	09:12:50	SELECT * FROM class CROSS JOIN cla...	36 row(s) returned	0.000 sec / 0.000 sec
21	09:13:58	SELECT * FROM class INNER JOIN cla...	9 row(s) returned	0.000 sec / 0.000 sec
22	09:14:18	SELECT class.name, class_info.add...	9 row(s) returned	0.000 sec / 0.000 sec
23	09:15:32	SELECT * FROM class NATURAL JOIN ...	9 row(s) returned	0.000 sec / 0.000 sec
24	09:15:49	INSERT INTO class_info VALUES (7, 'N...	2 row(s) affected Records: 2 Duplicates:...	0.015 sec
25	09:15:56	SELECT * FROM class LEFT OUTER J...	10 row(s) returned	0.015 sec / 0.000 sec

6. Natural Join Query

The screenshot shows the SQL Developer interface with a query window titled 'SQL File 3*' containing the following SQL code:

```

87 SELECT *
88 FROM class
89 LEFT JOIN class_info
90 ON class.id = class_info.id
91 WHERE class_info.id IS NULL;

```

The 'Result Grid' shows the following data:

id	name	id	address
4	anu	NULL	NULL

The 'Output' pane shows the 'Action Output' table with the following rows:

#	Time	Action	Message	Duration / Fetch
14	09:10:45	SHOW TABLES	2 row(s) returned	0.000 sec / 0.000 sec
15	09:11:02	INSERT INTO class VALUES (1, 'abhi'), ...	4 row(s) affected Records: 4 Duplicates:...	0.000 sec
16	09:11:10	SELECT * FROM class LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
17	09:12:04	CREATE TABLE class_info (id INT, ...	Error Code: 1050. Table 'class_info' alrea...	0.000 sec
18	09:12:20	INSERT INTO class_info VALUES (1, 'D...	3 row(s) affected Records: 3 Duplicates:...	0.031 sec
19	09:12:30	SELECT * FROM class_info LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
20	09:12:50	SELECT * FROM class CROSS JOIN cla...	36 row(s) returned	0.000 sec / 0.000 sec
21	09:13:58	SELECT * FROM class INNER JOIN cla...	9 row(s) returned	0.000 sec / 0.000 sec
22	09:14:18	SELECT class.name, class_info.add...	9 row(s) returned	0.000 sec / 0.000 sec
23	09:15:32	SELECT * FROM class NATURAL JOIN ...	9 row(s) returned	0.000 sec / 0.000 sec
24	09:15:49	INSERT INTO class_info VALUES (7, 'N...	2 row(s) affected Records: 2 Duplicates:...	0.015 sec
25	09:15:56	SELECT * FROM class LEFT OUTER J...	10 row(s) returned	0.015 sec / 0.000 sec
26	09:16:10	SELECT * FROM class LEFT JOIN class...	1 row(s) returned	0.000 sec / 0.000 sec

A tooltip on the right states: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

7. Left Outer Join Queries

The screenshot shows the SQL Developer interface with a query window titled 'SQL File 3*' containing the following SQL code:

```

92
93 SELECT *
94 FROM class
95 RIGHT OUTER JOIN class_info
96 ON class.id = class_info.id;

```

The 'Result Grid' shows the following data:

id	name	id	address
1	abhi	1	DELHI
2	adam	2	MUMBAI
3	alex	3	CHENNAI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI

The 'Output' pane shows the 'Action Output' table with the following rows:

#	Time	Action	Message	Duration / Fetch
15	09:11:02	INSERT INTO class VALUES (1, 'abhi'), ...	4 row(s) affected Records: 4 Duplicates:...	0.000 sec
16	09:11:10	SELECT * FROM class LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
17	09:12:04	CREATE TABLE class_info (id INT, ...	Error Code: 1050. Table 'class_info' alrea...	0.000 sec
18	09:12:20	INSERT INTO class_info VALUES (1, 'D...	3 row(s) affected Records: 3 Duplicates:...	0.031 sec
19	09:12:30	SELECT * FROM class_info LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
20	09:12:50	SELECT * FROM class CROSS JOIN cla...	36 row(s) returned	0.000 sec / 0.000 sec
21	09:13:58	SELECT * FROM class INNER JOIN cla...	9 row(s) returned	0.000 sec / 0.000 sec
22	09:14:18	SELECT class.name, class_info.add...	9 row(s) returned	0.000 sec / 0.000 sec
23	09:15:32	SELECT * FROM class NATURAL JOIN ...	9 row(s) returned	0.000 sec / 0.000 sec
24	09:15:49	INSERT INTO class_info VALUES (7, 'N...	2 row(s) affected Records: 2 Duplicates:...	0.015 sec
25	09:15:56	SELECT * FROM class LEFT OUTER J...	10 row(s) returned	0.015 sec / 0.000 sec
26	09:16:10	SELECT * FROM class LEFT JOIN class...	1 row(s) returned	0.000 sec / 0.000 sec
27	09:16:26	SELECT * FROM class RIGHT OUTER ...	11 row(s) returned	0.000 sec / 0.000 sec

A tooltip on the right states: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

8. Right Outer Join Queries

The screenshot shows the SQL Studio interface with a query editor and a results grid. The query is a Right Outer Join between 'class' and 'class_info' tables. The results grid shows two rows where 'class_info' has data but 'class' does not.

Query:

```

98 SELECT *
99 FROM
100 RIGHT JOIN class_info
101 ON class.id = class_info.id
102 WHERE class.id IS NULL;

```

Results Grid:

	id	name	id	address
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Output Log:

#	Time	Action	Message	Duration / Fetch
16	09:11:10	SELECT * FROM class LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
17	09:12:04	CREATE TABLE class_info (id INT, ...	Error Code: 1050. Table 'class_info' alrea...	0.000 sec
18	09:12:20	INSERT INTO class_info VALUES (1, 'D...	3 row(s) affected Records: 3 Duplicates:...	0.031 sec
19	09:12:30	SELECT * FROM class_info LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
20	09:12:50	SELECT * FROM class CROSS JOIN cla...	36 row(s) returned	0.000 sec / 0.000 sec
21	09:13:58	SELECT * FROM class INNER JOIN cla...	9 row(s) returned	0.000 sec / 0.000 sec
22	09:14:18	SELECT class.name, class_info.add...	9 row(s) returned	0.000 sec / 0.000 sec
23	09:15:32	SELECT * FROM class NATURAL JOIN ...	9 row(s) returned	0.000 sec / 0.000 sec
24	09:15:49	INSERT INTO class_info VALUES (7, 'N...	2 row(s) affected Records: 2 Duplicates:...	0.015 sec
25	09:15:56	SELECT * FROM class LEFT OUTER J...	10 row(s) returned	0.015 sec / 0.000 sec
26	09:16:10	SELECT * FROM class LEFT JOIN class...	1 row(s) returned	0.000 sec / 0.000 sec
27	09:16:26	SELECT * FROM class RIGHT OUTER ...	11 row(s) returned	0.000 sec / 0.000 sec
28	09:16:39	SELECT * FROM class RIGHT JOIN cla...	2 row(s) returned	0.000 sec / 0.000 sec

9. Full Outer Join Queries

The screenshot shows the SQL Studio interface with a query editor and a results grid. The query is a Full Outer Join between 'class' and 'class_info' tables. The results grid shows three rows: one from 'class' only, one from 'class_info' only, and one from both.

Query:

```

73 SELECT *
74 FROM class
75 LEFT JOIN class_info ON class.id = class_info.id
76 WHERE class_info.id IS NULL
77 UNION
78 SELECT *
79 FROM class
80 RIGHT JOIN class_info ON class.id = class_info.id
81 WHERE class.id IS NULL;

```

Results Grid:

	id	name	id	address
	4	anu	NULL	NULL
	5	ashish	NULL	NULL
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Output Log:

#	Time	Action	Message	Duration / Fetch
47	13:39:49	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
48	13:39:56	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
49	13:40:02	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT ...	2 row(s) returned	0.000 sec / 0.000 sec
50	13:40:08	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT ...	2 row(s) returned	0.000 sec / 0.000 sec
51	13:40:48	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, ...	2 row(s) returned	0.000 sec / 0.000 sec
52	13:41:27	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL UNIO...	4 row(s) returned	0.000 sec / 0.000 sec

10. Setup Tables for Set Operations

```

82 • CREATE TABLE first_table (
83     id INT,
84     name VARCHAR(30)
85 );
86
87 • CREATE TABLE second_table (
88     id INT,
89     name VARCHAR(30)
90 );
91
92 • INSERT INTO first_table VALUES
93     (1, 'abhi'),
94     (2, 'adam');
95
96 • INSERT INTO second_table VALUES
97     (2, 'adam'),
98     (3, 'chester');

```

Output

#	Time	Action	Message	Duration / Fetch
52	13:41:27	SELECT * FROM class LEFT JOIN class_info ON class_id = class_info_id WHERE class_info_id IS NULL UNIO...	4 row(s) returned	0.000 sec / 0.000 sec
53	13:42:06	INSERT INTO second_table VALUES (2, 'adam'), (3, 'chester')	Error Code: 1146 Table 'school_db second_table' doesn't exist	0.000 sec
54	13:42:14	CREATE TABLE first_table (id INT, name VARCHAR(30))	0 row(s) affected	0.000 sec
55	13:42:17	CREATE TABLE second_table (id INT, name VARCHAR(30))	0 row(s) affected	0.015 sec
56	13:42:21	INSERT INTO first_table VALUES (1, 'abhi'), (2, 'adam')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.015 sec
57	13:42:24	INSERT INTO second_table VALUES (2, 'adam'), (3, 'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec

11. Union Operations

```

99 • SELECT * FROM first_table
100 UNION
101 SELECT * FROM second_table;
102
103 • SELECT name FROM first_table
104 UNION
105 SELECT name FROM second_table;

```

Result Grid

id	name
1	abhi
2	adam
3	chester

Result 15

Output

#	Time	Action	Message	Duration / Fetch
55	13:42:17	CREATE TABLE second_table (id INT, name VARCHAR(30))	0 row(s) affected	0.015 sec
56	13:42:21	INSERT INTO first_table VALUES (1, 'abhi'), (2, 'adam')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.015 sec
57	13:42:24	INSERT INTO second_table VALUES (2, 'adam'), (3, 'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
58	13:42:57	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
59	13:43:03	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
60	13:43:10	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec

12. Union All Operations

```

109
110 • SELECT COUNT(*)
111 FROM (
112     SELECT * FROM first_table
113     UNION ALL
114     SELECT * FROM second_table
115 ) AS total_records;

```

Result Grid

COUNT(*)
4

Result 16

Output

#	Time	Action	Message	Duration / Fetch
56	13:42:21	INSERT INTO first_table VALUES (1, 'abhi'), (2, 'adam')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.015 sec
57	13:42:24	INSERT INTO second_table VALUES (2, 'adam'), (3, 'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
58	13:42:57	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
59	13:43:03	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
60	13:43:10	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
61	13:44:10	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table) AS total...	1 row(s) returned	0.000 sec / 0.000 sec

13. Intersect Operations

119 • `SELECT name FROM first_table`

Result Grid

id	name
2	adam

Result 23 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
64	13:44:52	<code>SELECT * FROM first_table INTERSECT SELECT * FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec
65	13:44:57	<code>SELECT * FROM first_table INTERSECT SELECT * FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec
66	13:45:03	<code>SELECT name FROM first_table INTERSECT SELECT name FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec
67	13:45:10	<code>SELECT name FROM first_table INTERSECT SELECT name FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec
68	13:45:13	<code>INTERSECT SELECT name FROM second_table</code>	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ser...	0.000 sec
69	13:45:36	<code>SELECT * FROM first_table INTERSECT SELECT * FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec

14. Minus / Except Operations

123 • `SELECT * FROM first_table`

124 `EXCEPT`

125 `SELECT * FROM second_table;`

126

127 • `SELECT name FROM first_table`

128 `EXCEPT`

129 `SELECT name FROM second_table;`

Result Grid

name
abhi

Result 26 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
71	13:46:37	<code>USE school_db</code>	0 row(s) affected	0.000 sec
72	13:46:37	<code>CREATE TABLE class (id INT, name VARCHAR(30))</code>	Error Code: 1050. Table 'class' already exists	0.000 sec
73	13:46:41	<code>SELECT name FROM first_table SELECT * FROM first_table EXCEPT SELECT * FROM second_table</code>	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ser...	0.000 sec
74	13:49:44	<code>SELECT name FROM first_table INTERSECT SELECT name FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec
75	13:49:49	<code>SELECT name FROM first_table INTERSECT SELECT name FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec
76	13:50:07	<code>SELECT name FROM first_table EXCEPT SELECT name FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec

15. Set Difference Alternative

129 `SELECT name FROM second_table;`

130 • `SELECT f.*`

131 `FROM first_table f`

132 `LEFT JOIN second_table s ON f.id = s.id`

133 `WHERE s.id IS NULL;`

134

135 • `SELECT f.name`

136 `FROM first_table f`

137 `LEFT JOIN second_table s ON f.name = s.name`

138 `WHERE s.name IS NULL;`

Result Grid

name
abhi

Result 30 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
75	13:49:49	<code>SELECT name FROM first_table INTERSECT SELECT name FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec
76	13:50:07	<code>SELECT name FROM first_table EXCEPT SELECT name FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec
77	13:50:53	<code>SELECT f.name FROM first_table f LEFT JOIN second_table s ON f.name = s.name WHERE s.name IS NULL ...</code>	1 row(s) returned	0.000 sec / 0.000 sec
78	13:50:58	<code>SELECT f.* FROM first_table f LEFT JOIN second_table s ON f.id = s.id WHERE s.id IS NULL LIMIT 0, 1000</code>	1 row(s) returned	0.000 sec / 0.000 sec
79	13:51:00	<code>SELECT f.* FROM first_table f LEFT JOIN second_table s ON f.id = s.id WHERE s.id IS NULL LIMIT 0, 1000</code>	1 row(s) returned	0.000 sec / 0.000 sec
80	13:51:05	<code>SELECT f.name FROM first_table f LEFT JOIN second_table s ON f.name = s.name WHERE s.name IS NULL ...</code>	1 row(s) returned	0.000 sec / 0.000 sec

16. Advanced Queries with Aliases and Case Expressions

```

144 • SELECT c.id,
145       c.name,
146       CASE
147       WHEN ci.address IS NULL THEN 'Address Missing'
148       ELSE 'Address Available'
149       END AS Status
150 FROM class c
151 LEFT JOIN class_info ci
152 ON c.id = ci.id;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	id	name	Status
▶	1	abhi	Address Available
	2	adam	Address Available
	3	alex	Address Available
	4	anu	Address Missing
	5	ashish	Address Missing

Result Grid
Form Editor

Result 31 x

Read Only

Output
Action Output

#	Time	Action	Message	Duration / Fetch
✓ 76	13:50:07	SELECT name FROM first_table EXCEPT SELECT name FROM second_table	1 row(s) returned	0.000 sec / 0.000 sec
✓ 77	13:50:53	SELECT f.name FROM first_table f LEFT JOIN second_table s ON f.name = s.name WHERE s.name IS NULL ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 78	13:50:58	SELECT f.* FROM first_table f LEFT JOIN second_table s ON f.id = s.id WHERE s.id IS NULL LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
✓ 79	13:51:00	SELECT f.* FROM first_table f LEFT JOIN second_table s ON f.id = s.id WHERE s.id IS NULL LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
✓ 80	13:51:05	SELECT f.name FROM first_table f LEFT JOIN second_table s ON f.name = s.name WHERE s.name IS NULL ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 81	13:51:58	SELECT c.id, c.name, CASE WHEN ci.address IS NULL THEN 'Address Missing' ELSE 'Address Available' EN...	5 row(s) returned	0.000 sec / 0.000 sec